Formal Definition of the Limit

$\varepsilon - \delta$ definition of a limit

If f is a function defined on an interval containing $x = c$, except possibly at $x = c$, then the limit of $f(x)$ as x approaches c is L. That is

$$\lim_{x \to c} f(x) = L$$

if for every number $\varepsilon > 0$ there exists a number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ when } 0 < |x - c| < \delta$$

or

$$\text{If } 0 < |x - c| < \delta, \text{ then } |f(x) - L| < \varepsilon$$

Exercises

1. Prove that $\lim_{x \to 1}(7x + 4) = 11$.

$|f(x) - L| < \varepsilon$ for $|x - 1| < \delta$

Guess

$|7x + 4 - 11| < \varepsilon$

$|7x - 7| < \varepsilon$

$7|x - 1| < \varepsilon$

$|x - 1| < \frac{\varepsilon}{7}$

so $\delta = \frac{\varepsilon}{7}$

X Find δ as a function of ε

proof:

if $\delta = \frac{\varepsilon}{7}$

then $|x - 1| < \delta$

$7|x - 1| < \frac{\varepsilon}{7} \cdot 7$

$|7x - 7| < \varepsilon$

$|7x + 4 - 11| < \varepsilon$

$|(7x + 4) - 11| < \varepsilon$

$|f(x) - L| < \varepsilon$ ✓

2. (1969 BC-37) Of the following choices of δ, which is the largest that could be used successfully with an arbitrary ε in an epsilon-delta proof of $\lim_{x \to 2}(1 - 3x) = -5$?

(A) $\delta = 3\varepsilon$

(B) $\delta = \varepsilon$

(C) $\delta = \frac{\varepsilon}{2}$

(D) $\delta = \frac{\varepsilon}{4}$

(E) $\delta = \frac{\varepsilon}{5}$

$|f(x) - L| < \varepsilon$ for $|x - 2| < \delta$

$|1 - 3x - (-5)| < \varepsilon$

$|6 - 3x| < \varepsilon$

$|-3(x - 2)| < \varepsilon$

$3|x - 2| < \varepsilon$

$|x - 2| < \frac{\varepsilon}{3}$

so $\delta = \frac{\varepsilon}{3}$ (or smaller)

Of the answer choices, only (D) $\delta = \frac{\varepsilon}{4} \leq \frac{\varepsilon}{3}$

3. Find the limit, then prove it: $\lim_{x\to -1}\left(\frac{2}{3}-\frac{1}{5}x\right)$ $= \frac{2}{3}-\frac{1}{5}(-1) = \frac{2}{3}+\frac{1}{5} = \frac{13}{15}$

$|f(x)-L|<\varepsilon$ for $|x-(-1)|<\delta$

$|x+1|<\delta$

$\left|\frac{2}{3}-\frac{1}{5}x-\frac{13}{15}\right|<\varepsilon$

$15\cdot\left|-\frac{1}{5}x-\frac{3}{15}\right|<\varepsilon\cdot 15$

$|-3x-3|<15\varepsilon$

$|-3(x+1)|<15\varepsilon$

$3|x+1|<15\varepsilon$

$|x+1|<5\varepsilon$

so $\delta = 5\varepsilon$

proof:

if $\delta = 5\varepsilon$

$\left(\frac{1}{3}\right)|x+1|<5\varepsilon\left(\frac{1}{3}\right)$

$\left|\frac{1}{3}x+\frac{1}{3}\right|<\varepsilon$

$\left|-\frac{1}{5}x-\frac{1}{5}\right|<\varepsilon$

$\left|-\frac{1}{5}x-\frac{3}{15}+\frac{2}{3}-\frac{2}{3}\right|<\varepsilon$

$\left|\left(\frac{2}{3}-\frac{1}{5}x\right)-\frac{3}{15}-\frac{10}{15}\right|<\varepsilon$

$\left|\left(\frac{2}{3}-\frac{1}{5}x\right)-\frac{13}{15}\right|<\varepsilon$

$|f(x)-L|<\varepsilon$ ✓



4. b) (4 points) Use the graph of $f(x) = 2\sqrt{x+1}$ to find a number $\delta > 0$ such that for all x,

$$0 < |x-3| < \delta \Rightarrow |f(x)-4| < 0.2$$

5. (8 - points) Given that $\lim_{x\to 2}\left(3x-\frac{2}{5}\right) = \frac{28}{5}$ and $\epsilon = 0.009$. Find the largest possi value of δ that satisfies the conditions given in the $\epsilon-\delta$ definition of a limit.

Derivative

Differentiation

The Chain Rule

If $y = f(u)$ and $u = g(x)$ are both differentiable functions, then $y = f(g(x))$ is differentiable and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

or, equivalently,

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x))g'(x).$$

The Power Chain Rule

If n is any real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}$$

or, equivalently,

$$\frac{d}{dx}\left[g(x)\right]^n = n\left[g(x)\right]^{n-1} \cdot g'(x).$$

Exercises

1. If $f(x) = \sqrt{x+\sqrt{x}}$, then $f'(x) =$

(A) $\frac{1}{2\sqrt{x+\sqrt{x}}}$ (B) $\frac{\sqrt{x}+1}{2\sqrt{x+\sqrt{x}}}$ (C) $\frac{2\sqrt{x}}{4\sqrt{x+\sqrt{x}}}$ (D) $\frac{2\sqrt{x}+1}{4\sqrt{x^2+x\sqrt{x}}}$

2. If $f(x) = (x^2 - 3x)^{3/2}$, then $f'(4) =$

(A) $\frac{15}{2}$ (B) 9 (C) $\frac{21}{2}$ (D) 15

3. If f, g, and h are functions that is everywhere differentiable, then the derivative of $\frac{f}{g \cdot h}$ is

(A) $\frac{g\,h\,f' - f\,g'\,h'}{g\,h}$

(B) $\frac{g\,h\,f' - f\,g\,h' - f\,h\,g'}{g\,h}$

(C) $\frac{g\,h\,f' - f\,g\,h' - f\,g'h}{g^2h^2}$

(D) $\frac{g\,h\,f' - f\,g\,h' + f\,h\,g'}{g^2h^2}$

4. Let f and g be differentiable functions such that $f(g(x)) = 2x$ and $f'(x) = 1 + [f(x)]^2$.

(a) Show that $g'(x) = \dfrac{2}{f'(g(x))}$.

(b) Show that $g'(x) = \dfrac{2}{1+4x^2}$.

5. Find $h'(1)$, if $h(x) = (x^9 + f(x))^{-2}$.

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

Note: $\sec^2 x$ means $(\sec x)^2$.

Exercises

1. $\lim_{h\to 0} \dfrac{\sin 2(x+h) - \sin 2x}{h} =$

(A) $2\sin 2x$ (B) $-2\sin 2x$ (C) $2\cos 2x$ (D) $-2\cos 2x$

2. If $f(x) = \sin(\cos 2x)$, then $f'(\frac{\pi}{4}) =$

(A) 0 (B) -1 (C) 1 (D) -2

3. If $y = a\sin x + b\cos x$, then $y + y'' =$

(A) 0 (B) $2a\sin x$ (C) $2b\cos x$ (D) $-2a\sin x$

4. If $f(\theta) = \cos\pi - \dfrac{1}{2\cos\theta} + \dfrac{1}{3\tan\theta}$, then $f'(\frac{\pi}{6}) =$

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{4}{\sqrt{3}}$ (D) $2\sqrt{3}$

The Derivatives of Exponential Function

$$\frac{d}{dx}\left[e^x\right]=e^x \qquad \frac{d}{dx}\left[e^u\right]=e^u\frac{du}{dx}$$

$$\frac{d}{dx}\left[a^x\right]=a^x\ln a \qquad \frac{d}{dx}\left[a^u\right]=a^u(\ln a)\frac{du}{dx}$$

The Derivatives of Logarithmic Function

$$\frac{d}{dx}(\ln x)=\frac{1}{x} \qquad \frac{d}{dx}(\log_a x)=\frac{1}{\ln a}\cdot\frac{1}{x}$$

$$\frac{d}{dx}(\ln u)=\frac{1}{u}\frac{du}{dx} \qquad \frac{d}{dx}(\log_a u)=\frac{1}{\ln a}\cdot\frac{1}{u}\frac{du}{dx}$$

Properties of Logarithms

$$\ln xy=\ln x+\ln y \qquad \ln\frac{x}{y}=\ln x-\ln y$$

$$\ln x^p=p\ln x \qquad e^{\ln x}=x$$

Exercises

1. $\lim_{h\to 0}\dfrac{\frac{1}{2}[\ln(e+h)-1]}{h}$ is

(A) $f'(1)$, where $f(x)=\ln\sqrt{x}$

(B) $f'(1)$, where $f(x)=\ln\sqrt{x+e}$

(C) $f'(e)$, where $f(x)=\ln\sqrt{x}$

(D) $f'(e)$, where $f(x)=\ln(\frac{x}{2})$

2. If $f(x)=e^{\tan x}$, then $f'(\frac{\pi}{4})=$

(A) $\frac{e}{2}$ (B) e (C) $2e$ (D) $\frac{e^2}{2}$

3. If $y=\ln(\cos x)$, then $y'=$

(A) $-\tan x$ (B) $\tan x$ (C) $-\cot x$ (D) $\csc x$

4. If $y = x^x$, then $y' =$

(A) $x^x \ln x$ (B) $x^x(1+\ln x)$ (C) $x^x(x+\ln x)$ (D) $\dfrac{x^x \ln x}{x}$

5. If $y = e^{\sqrt{x^2+1}}$, then $y' =$

(A) $\sqrt{x^2+1}\, e^{\sqrt{x^2+1}}$

(B) $2x\sqrt{x^2+1}\, e^{\sqrt{x^2+1}}$

(C) $\dfrac{e^{\sqrt{x^2+1}}}{\sqrt{x^2+1}}$

(D) $\dfrac{xe^{\sqrt{x^2+1}}}{\sqrt{x^2+1}}$

6. If $f(x) = \ln[\sec(\ln x)]$, then $f'(e) =$

(A) $\dfrac{\cos 1}{e}$ (B) $\dfrac{\sin 1}{e}$ (C) $\dfrac{\tan 1}{e}$ (D) $\dfrac{\cot 1}{e}$

7. Let f and h be twice differentiable functions such that $h(x) = e^{f(x)}$. If $h''(x) = e^{f(x)}[1$ then $f'(x) =$

The **slope of the tangent line** to the graph of f at the point $P(a, f(a))$ is the number

$$m = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

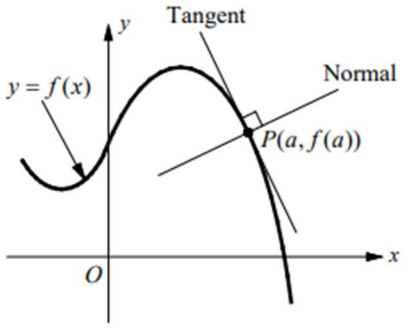
if the limit exists.

Recall from the definition of the derivative that this limit is just $f'(a)$.

In **point slope form** the equation of the tangent line is

$$y - f(a) = f'(a)(x - a)$$

The **normal line** to the graph of f at the point $P(a, f(a))$ is the line that passes through P and is perpendicular to the tangent line to the graph of f at P.



Since the tangent line and the normal line are perpendicular, their slopes are negative reciprocal of each other. Therefore $m_1 \cdot m_2 = -1$.

Exercises

1. The equation of the line tangent to the graph of $y = x\sqrt{3+x^2}$ at the point $(1,2)$ is

(A) $y = \frac{3}{2}x - \frac{1}{2}$ (B) $y = 2x + \frac{1}{2}$ (C) $y = \frac{5}{2}x - \frac{1}{2}$ (D) $y = \frac{5}{2}x + \frac{1}{2}$

2. A curve has slope $2x + x^{-2}$ at each point (x, y) on the curve. Which of the following is an equation fo this curve if it passes through the point $(1,3)$?

(A) $y = 2x^2 + \frac{1}{x}$

(B) $y = x^2 - \frac{1}{x} + 3$

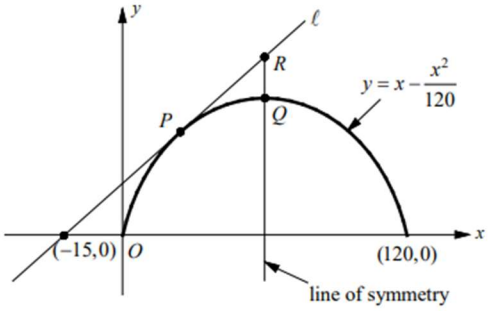
(C) $y = x^2 + \frac{1}{x} + 1$

(D) $y = x^2 - \frac{2}{x^2} + 4$

3. If $2x + 3y = 4$ is an equation of the line normal to the graph of f at the point $(-1,2)$, then $f'(-1) =$

(A) $-\frac{2}{3}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\sqrt{2}$ (D) $\frac{3}{2}$

4.



Line ℓ is tangent to the graph of $y = x - \frac{x^2}{120}$ at the point P and intersects x-axis at $(-15,0)$ as shown in the figure above.

(a) Find the x-coordinates of point P.

(b) Write an equation for line ℓ.

(c) If the line of symmetry for the curve $y = x - \frac{x^2}{120}$ intersects line ℓ at point R, what is the length of $\overline{QR}$?

Implicit Differentiation

The functions that we have met so far have been described by an equation, of the form $y = f(x)$, that expresses y explicitly in terms of x for example, $y = \frac{1}{x^2+1}$ or $y = x + \cos x$. Some functions, however, are defined **implicitly** by a relation between x and y such as $x^2 + 4y^2 - 9 = 0$ or $\sin xy = x^2 + 1$. In some cases it is possible to solve such an equation for y as an explicit function of x. But when we are unable to solve for y as a function of x, we may still be able to find $\frac{dy}{dx}$ by using the method of **implicit differentiation**.

Guidelines for Implicit Differentiation

1. Differentiate both sides of the equation with respect to x.
2. Collect the terms with dy/dx on the left side of the equation and move all other terms to the right side of the equation.
3. Solve for dy/dx.

The **tangent line is horizontal** if $dy/dx = 0$.
The **tangent line is vertical** when the denominator in the expression for dy/dx is 0.

Exercises

1. If $3xy + x^2 - 2y^2 = 2$, then the value of $\frac{dy}{dx}$ at the point $(1,1)$ is

(A) 5 (B) $\frac{7}{2}$ (C) $-\frac{1}{2}$ (D) $-\frac{7}{2}$

2. If $3x^4 - x^2 - y^2 = 0$, then the value of $\frac{dy}{dx}$ at the point $(1,\sqrt{2})$ is

(A) $\frac{\sqrt{2}}{2}$ (B) $\frac{3\sqrt{2}}{2}$ (C) $\frac{5\sqrt{2}}{2}$ (D) $\frac{7\sqrt{2}}{2}$

3. If $xy + \tan(xy) = \pi$, then $\frac{dy}{dx} =$

(A) $-y\sec^2(xy)$ (B) $-y\cos^2(xy)$ (C) $-x\sec^2(xy)$ (D) $-\frac{y}{x}$

4. If $x + \sin y = y + 3$, then $\frac{d^2y}{dx^2} =$

(A) $\frac{-\sin y}{(1-\cos y)^2}$ (B) $\frac{-\sin y}{(1+\cos y)^2}$ (C) $\frac{-\sin y}{(1-\cos y)^3}$ (D) $\frac{-\sin y}{(1+\cos y)^3}$

5. Consider the curve given by $x^3 - xy + y^2 = 3$.

(a) Find $\frac{dy}{dx}$.

(b) Find all points on the curve whose x-coordinate is 1, and write an equation for the tangent li
each of these points.

(c) Find the x-coordinate of each point on the curve where the tangent line is horizontal.

Derivative of an Inverse Function

The Derivative of an Inverse Function

Let f be a differentiable function whose inverse function f^{-1} is also differentiable. Then, providing that the denominator is not zero,

$$(f^{-1})'(x) = \frac{1}{f'\left(f^{-1}(x)\right)} \quad \text{or} \quad (f^{-1})'(a) = \frac{1}{f'\left(f^{-1}(a)\right)}.$$

Exercises

1. Let f and g be functions that are differentiable everywhere. If g is the inverse function of f and if $g(3) = 4$ and $f'(4) = \frac{3}{2}$, then $g'(3) =$

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{4}{3}$

2. If $f(-3) = 2$ and $f'(-3) = \frac{3}{4}$, then $(f^{-1})'(2) =$

(A) $\frac{1}{2}$ (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{4}$

3. If $f(x) = \sin x$, then $(f^{-1})'(\frac{\sqrt{3}}{2}) =$

(A) $\frac{1}{2}$ (B) $\frac{2\sqrt{3}}{3}$ (C) $\sqrt{3}$ (D) 2

4. If $f(x) = 1 + \ln x$, then $(f^{-1})'(2) =$

(A) $-\frac{1}{e}$ (B) $\frac{1}{e}$ (C) $-e$ (D) e

5.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-1	3	-2	2	6
0	-2	-1	0	-3
1	0	1	-1	2
2	-1	4	3	-1

The functions f and g are differentiable for all real numbers. The table above gives the values of the functions and their first derivatives at selected values of x.

(a) If f^{-1} is the inverse function of f, write an equation for the line tangent to the graph of $y = f^{-1}(x)$ at $x = -1$.

(b) Let h be the function given by $h(x) = f(g(x))$. Find $h(1)$ and $h'(1)$.

(c) Find $(h^{-1})'(3)$, if h^{-1} is the inverse function of h.

Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}(\sin^{-1}x)=\frac{1}{\sqrt{1-x^2}} \qquad \frac{d}{dx}(\cos^{-1}x)=-\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1}x)=\frac{1}{1+x^2} \qquad \frac{d}{dx}(\cot^{-1}x)=-\frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sec^{-1}x)=\frac{1}{x\sqrt{x^2-1}} \qquad \frac{d}{dx}(\csc^{-1}x)=-\frac{1}{x\sqrt{x^2-1}}$$

Exercises

1. $\frac{d}{dx}(\arcsin x^2)=$

(A) $-\frac{2x}{\sqrt{1-x^2}}$ (B) $\frac{2x}{\sqrt{x^2-1}}$ (C) $\frac{2x}{\sqrt{x^4-1}}$ (D) $\frac{2x}{\sqrt{1-x^4}}$

2. If $f(x)=\arctan(e^{-x})$, then $f'(-1)=$

(A) $\frac{-e}{1+e}$ (B) $\frac{e}{1+e}$ (C) $\frac{-e}{1+e^2}$ (D) $\frac{-1}{1+e^2}$

3. If $f(x)=\arctan(\sin x)$, then $f'(\frac{\pi}{3})=$

(A) $\frac{2}{7}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{2}}{3}$ (D) $\frac{\sqrt{3}}{3}$

4. If $y=\cos(\sin^{-1}x)$, then $y'=$

(A) $-\frac{1}{\sqrt{1-x^2}}$ (B) $-\frac{x}{\sqrt{1-x^2}}$ (C) $\frac{2x}{\sqrt{1-x^2}}$ (D) $-\frac{2x}{\sqrt{x^2-1}}$

5. Let f be the function given by $f(x)=x^{\tan^{-1}x}$.

(a) Find $f'(x)$.

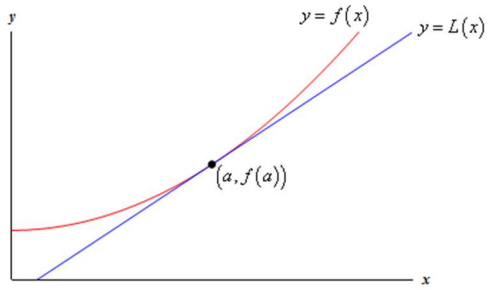
(b) Write an equation for the line tangent to the graph of f at $x=1$.

Linearization

Given a function, $f(x)$, we can find its tangent at $x = a$. The equation of the tangent line, which we'll call $L(x)$ for this discussion, is,

$$L(x) = f(a) + f'(a)(x - a)$$

Take a look at the following graph of a function and its tangent line.



From this graph we can see that near $x = a$ the tangent line and the function have nearly the same graph. On occasion we will use the tangent line, $L(x)$, as an approximation to the function, $f(x)$, near $x = a$. In these cases we call the tangent line the **linear approximation** to the function at $x = a$.

Exercises

Example 1 Determine the linear approximation for $f(x) = \sqrt[3]{x}$ at $x = 8$. Use the linear approximation to approximate the value of $\sqrt[3]{8.05}$ and $\sqrt[3]{25}$.

3. Find the linear approximation to $g(z) = \sqrt[4]{z}$ at $z = 2$. Use the linear approximation to approximate the value of $\sqrt[4]{3}$ and $\sqrt[4]{10}$. Compare the approximated values to the exact values. **[Solution]**

Example 2 Determine the linear approximation for $\sin\theta$ at $\theta = 0$.

If $L(x)$ is the linearization of $f(x) = 2 + \ln(1 - 4x)$ at $a = 0$, then $L(-1) =$

Exercise 1. Use linear approximation to approximate $(1.001)^{1000}$

Related Rates

In a related rates problem, the idea is to compute the rates of change of two or more related variables with respect to time. The procedure involves finding an equation that relates the variables and using the Chain Rule to differentiate both sides with respect to time.

Guidelines For Solving Related Rate Problems

1. Read the problem carefully and draw a diagram if possible.
2. Name the variables and constants. Use t for time.
3. Write an equation that relates the variables whose rates of change are given. If necessary, use the geometry or trigonometry of the situation and write an equation whose rates of change are to be determined.
4. Combine two or more equations to get a single equation that relates the variable.
5. Use the Chain Rule to differentiate both sides of the equation with respect to t.
6. Substitute the given numerical information into the resulting equation and solve for the unknown rate.

Exercises

1. The radius of a circle is changing at the rate of $1/\pi$ inches per second. At what rate, in square inches per second, is the circle's area changing when $r = 5$ in ?

(A) $\frac{5}{\pi}$ (B) 10 (C) $\frac{10}{\pi}$ (D) 15

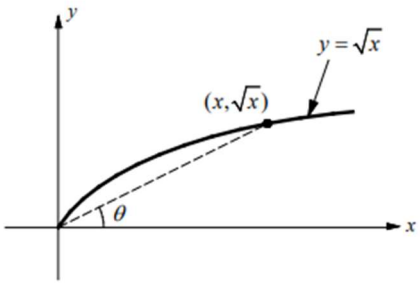
2. The volume of a cube is increasing at the rate of 12 in^3/min . How fast is the surface area increasing, in square inches per minute, when the length of an edge is 20 in?

(A) 1 (B) $\frac{6}{5}$ (C) $\frac{4}{3}$ (D) $\frac{12}{5}$

3. A car is approaching a right-angled intersection from the north at 70 mph and a truck is traveling to the east at 60 mph. When the car is 1.5 miles north of the intersection and the truck is 2 miles to the east, at what rate, in miles per hour, is the distance between the car and truck is changing?

(A) Decreasing 15 miles per hour
(B) Decreasing 9 miles per hour
(C) Increasing 6 miles per hour
(D) Increasing 12 miles per hour

4.



A particle moves along the curve $y = \sqrt{x}$. When $y = 2$ the x-component of its position is increasing at the rate of 4 units per second.

(a) What is the value of $\frac{dy}{dt}$ when $y = 2$?

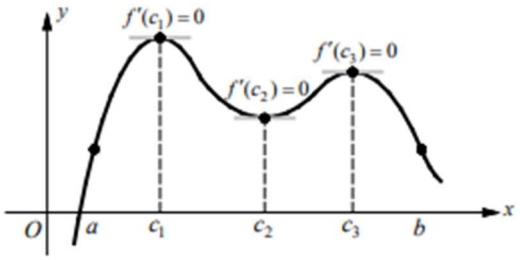
(b) How fast is the distance from the particle to the origin changing when $y = 2$?

(c) What is the value of $\frac{d\theta}{dt}$ when $y = 2$?

Rolle's Theorem

Let f be continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b). If $f(a)=f(b)$, then there is at least one number c in the open interval (a,b) such that

$$f'(c)=0.$$

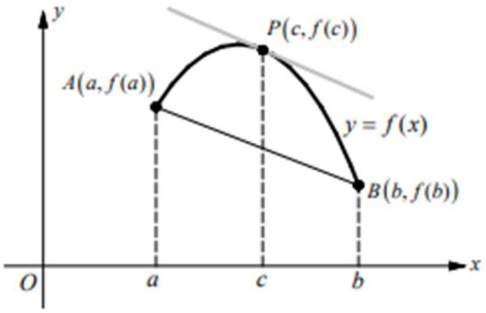


Rolle's Theorem says that a differentiable curve has at least one horizontal tangent between a and b if $f(a)=f(b)$.

The Mean Value Theorem

If f is continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b), then there exists a number c in the open interval (a,b) such that

$$f'(c)=\frac{f(b)-f(a)}{b-a}.$$



The Mean Value Theorem says that there is at least one point $P(c,f(c))$, on the curve where the tangent line is parallel to the secant line AB.

Exercises

1. Let f be the function given by $f(x)=\sin(\pi x)$. What are the values of c that satisfy Roll's Theorem on the closed interval $[0,2]$?

(A) $\frac{1}{4}$ only (B) $\frac{1}{2}$ only (C) $\frac{1}{4}$ and $\frac{1}{2}$ (D) $\frac{1}{2}$ and $\frac{3}{2}$

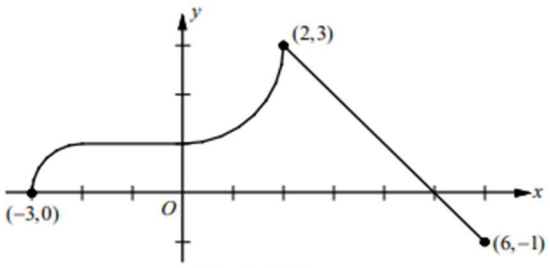
2. Let f be the function given by $f(x)=-x^3+3x+2$. What are the values of c that satisfy Mean Value Theorem on the closed interval $[0,3]$?

(A) $-\sqrt{3}$ only (B) $-\sqrt{3}$ and $\sqrt{3}$ (C) $\sqrt{3}$ only (D) 1.5 and $\sqrt{3}$

3. Let f be the function given by $f(x)=\dfrac{x}{x+2}$. What are the values of c that satisfy the Mean Value Theorem on the closed interval $[-1,2]$?

(A) -4 only (B) 0 only (C) 0 and $\dfrac{3}{2}$ (D) -4 and 0

4.



Graph of f

The continuous function f is defined on the interval $-3\le x\le 6$. The graph of f consists of two quarter circles and two line segments, as shown in the figure above. Which of the following statements must be true?

I. The average rate of change of f on the interval $-3\le x\le 6$ is $-\dfrac{1}{9}$.

II. There is a point c on the interval $-3<x<6$, for which $f'(c)$ is equal to the average rate of change of f on the interval $-3\le x\le 6$.

III. If h is the function given by $h(x)=f(\dfrac{1}{2}x)$, then $h'(6)=-\dfrac{1}{2}$.

(A) I and II only

(B) I and III only

(C) II and III only

(D) I, II, and III